

An Auto-Ranging Capacitance Meter on the ATmega328P

Charge-time measurement, supply-independent dual-threshold timing, and Bluetooth telemetry

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A digital capacitance meter is built on an ATmega328P. An unknown capacitor is charged through a known resistor and the time taken to cross two supply-referenced thresholds is timed in hardware; choosing the thresholds at one-third and two-thirds of the rail makes the result independent of supply voltage. A 16-channel analog multiplexer gives automatic ranging over seven decades, and a two-point calibration removes the residual systematic error. Measurements from 100 pF to 470 µF are presented, and the dominant low-range limit — multiplexer on-resistance — is identified and quantified.

1 Introduction

A microcontroller cannot read capacitance directly, but it can **measure time** with great precision. This instrument therefore converts the unknown capacitance into a time interval — using nothing more exotic than a resistor charging a capacitor — and counts that interval with a hardware timer. The design’s value lies in arranging the measurement so the result depends only on quantities we know well (a resistor and a clock) and not on those we do not (the exact supply, the comparator’s switching point, or temperature). We derive the measurement equation (§2), motivate automatic ranging mathematically and physically (§3), describe the architecture (§4) and calibration (§5), and present and discuss results (§6–7).

2 The measurement principle

2.1 The RC step response

Consider a capacitor C charged from a constant source V_{cc} through a series resistor R . The node voltage $v(t)$ obeys the first-order linear differential equation

$$RC \frac{dv}{dt} + v(t) = V_{cc},$$

whose solution, for an initially discharged capacitor, is the well-known exponential approach

$$v(t) = V_{cc} (1 - e^{-t/RC}). \quad (1)$$

The product $\tau = RC$ is the **time constant**: the time for the node to reach $1 - e^{-1} \approx 63.2\%$ of V_{cc} . Because the output is taken *across the capacitor*, the network is a first-order **low-pass filter** — the capacitor voltage cannot change instantaneously, so it integrates the input and yields the smooth curve of Equation 1. (Taking the output across the resistor would give a high-pass response — a decaying spike — which carries no usable timing information for our purpose.)

2.2 Inverting the curve to recover capacitance

Rearranging Equation 1, the time taken to reach a fraction of the supply is

$$t = -RC \ln \left(1 - \frac{v}{V_{cc}} \right). \quad (2)$$

A single-threshold scheme would charge from zero and stop the clock when v crosses one reference. This works, but Equation 2 shows that the result then depends explicitly on V_{cc} through the ratio v/V_{cc} — any drift or noise in the supply propagates directly into the capacitance.

2.3 Removing the supply dependence: the two-threshold method

We instead measure the *interval* between two thresholds, V_{low} and V_{high} , crossed as the node charges. Applying Equation 2 at each threshold and subtracting,

$$\Delta t = t_{\text{high}} - t_{\text{low}} = RC \ln \frac{V_{cc} - V_{\text{low}}}{V_{cc} - V_{\text{high}}}. \quad (3)$$

Now choose the thresholds as fixed fractions of the supply, generated by a resistor divider so that they *track* V_{cc} :

$$V_{\text{low}} = \frac{1}{3} V_{cc}, \quad V_{\text{high}} = \frac{2}{3} V_{cc}.$$

Substituting into Equation 3, the supply cancels completely:

$$\Delta t = RC \ln \frac{V_{cc} - \frac{1}{3} V_{cc}}{V_{cc} - \frac{2}{3} V_{cc}} = RC \ln \frac{2/3}{1/3} = RC \ln 2. \quad (4)$$

The capacitance is therefore

$$C = \frac{\Delta t}{R \ln 2} = \frac{\Delta t}{0.693 R} \quad (5)$$

This is a remarkably clean result. The measured capacitance depends only on the **known resistor**, the **measured time interval**, and a **dimensionless constant** $\ln 2$. The supply voltage, which is the least stable quantity in a hobby circuit, has vanished. These particular thresholds — one-third and two-thirds of the rail — will be familiar: they are the same levels used internally by the classic 555 timer, and for the same reason.

3 Why automatic ranging is necessary

Equation 5 suggests that *any* resistor would do. In practice a fixed resistor measures only a narrow band of capacitance, for two independent reasons — one arising from the digital timer, the other from the physics of the RC circuit.

3.1 The mathematical argument: a bounded counter

The interval Δt is not measured continuously but counted in discrete timer ticks of duration t_{tick} :

$$N = \frac{\Delta t}{t_{\text{tick}}}, \quad N_{\min} \leq N \leq N_{\max}.$$

Both bounds are real. At the **lower** end, the quantization error is ± 1 tick, i.e. a *relative* error of $\pm 1/N$; a count of $N = 5$ carries a 20% uncertainty, so we require $N \gtrsim 200$ for sub-percent resolution. At the **upper** end, the 16-bit timer saturates at $N_{\max} = 65535$; beyond this the counter overflows and the reading is meaningless.

A single resistor therefore admits only the capacitance band

$$C \in \left[\frac{N_{\min}}{0.693 R} t_{\text{tick}}, \frac{N_{\max}}{0.693 R} t_{\text{tick}} \right],$$

a span of $N_{\max}/N_{\min} \approx 65535/200 \approx 330$, or about **2.5 decades**. To cover the design target of 1 pF–1 μ F (six decades) — let alone the picofarad-to-millifarad range the hardware actually reaches — one resistor is mathematically insufficient. Since $\Delta t \propto RC$, keeping Δt inside the good window as C varies requires scaling R *inversely* with C . Because C is unknown a priori, the instrument must **try resistors until the count lands in range** — this is automatic ranging. A geometric ladder of resistors, each a fixed multiple of the last, tiles the whole span with overlapping windows.

3.2 The physical argument: matching the timescale

The same conclusion follows from physics. The time constant $\tau = RC$ sets how quickly the capacitor charges. If τ is *too short* (small RC) the node reaches the thresholds faster than the comparator's propagation delay and the timer's resolution can follow — one would be timing electrical artefacts, not the capacitor. If τ is *too long*, a measurement takes seconds to minutes and the counter overflows.

Every instrument must arrange for the physical event it measures to unfold on a timescale its sensor can resolve. Auto-ranging selects R so that charging occupies a *comfortable* window — long enough to accumulate many counts (good resolution), short enough to finish quickly and not overflow. The principle is identical to a multimeter's range switch (200 mV / 2 V / 20 V ...) or to choosing appropriate units on a stopwatch when timing a sprinter versus a glacier: one matches the scale of the tool to the magnitude of the quantity, so that the full resolution is used without running off the end of the scale.

4 Instrument architecture

Cx	R(range)	node	analog comparator	Timer1 input capture
	CD4067 mux (auto-range R)		1/3 & 2/3 Vcc divider	
7-seg display	SPI/595		HC-06	USART (Bluetooth telemetry)

Timing front-end. The node drives the on-chip analog comparator's positive input; the negative input is switched between the two reference taps through the ADC multiplexer, so the threshold is changed by a single register write. The comparator output is routed internally to the Timer1 *input-capture* unit, which latches the count in hardware at each crossing, free of interrupt-latency jitter.

Range selection and references. A CD74HC4067 16-channel multiplexer places one of a ladder of resistors between the charge pin and the node, selected by four digital lines. Three equal resistors from V_{cc} to ground form the $\frac{1}{3}V_{cc}$ and $\frac{2}{3}V_{cc}$ taps; since these and the charge source share the rail, the supply cancels (Equation 4) and the divider ratio need not even be precise — calibration absorbs it.

Display and telemetry. A four-digit seven-segment display is multiplexed through 74-series shift registers over hardware SPI, and results are streamed as ASCII over UART to an HC-06 Bluetooth module, which also lets a host trigger measurements remotely for collecting the calibration dataset below.

5 Calibration methodology

5.1 The need for, and form of, calibration

Equation 5 is the *ideal* relationship. The realized instrument departs from it through several systematic effects: the internal RC oscillator's frequency (hence t_{tick}) is accurate only to about $\pm 10\%$; the reference ratio is not exactly $\ln 2$; the resistors carry tolerance; the multiplexer adds series resistance; and the comparator and stray capacitance introduce small offsets. Rather than measure each effect separately, we absorb them all by fitting the instrument's response to capacitors of *known* value.

The multiplicative effects (clock, ratio, resistance) scale the slope, while the additive effects (stray capacitance, comparator delay) contribute an offset. The appropriate model is therefore an **affine** relation,

$$C = a \Delta t + b, \tag{6}$$

with slope a (in pF per tick) and intercept b (in pF).

5.2 Two-point calibration

Two known capacitors $(\Delta t_1, C_1)$ and $(\Delta t_2, C_2)$ determine the line uniquely:

$$a = \frac{C_2 - C_1}{\Delta t_2 - \Delta t_1}, \quad b = C_1 - a \Delta t_1. \quad (7)$$

A *single* point would fix only the slope and force the line through the origin ($b = 0$); we shall see in Section 6 that this is inadequate, because a non-zero offset is present. Two points capture both parameters, so the calibrated line is correct across the whole range rather than at one capacitor only.

5.3 A shortcut from measured resistors

Because each ladder resistor is individually measured, the per-range slope need not be calibrated separately. Writing $a = K/R$ with a single global constant $K = t_{\text{tick}}/\ln 2$, the model becomes

$$C = \frac{\Delta t K}{R} + b, \quad (8)$$

so that calibrating K and b *once* defines every range, each range's slope following from its measured resistance. This collapses a sixteen-range calibration into the determination of two numbers.

6 Results

6.1 Single-range characterization

The range resistor was first locked to channel 7 ($R = 99.3 \text{ k}\Omega$) and a set of known capacitors measured, each as the trimmed mean of 30 charge cycles (Table 1).

Table 1: Channel-7 characterization. The usable window (3–90 nF) agrees with the predicted span of Section 3.

Capacitor	Δt (ticks)	Within window?
100 pF	12	No — too fast ($\pm 8\%$ quantization)
10 nF (mylar)	733	Yes
“104” ceramic	2802	Yes ($\approx 40 \text{ nF}$; DC-bias loss)
50 nF	3579	Yes
100 nF	7788	Yes
1 μF	overflow	No — too slow

Applying the two-point fit of Equation 7 to the 10 nF and 100 nF points,

$$a = \frac{100000 - 10000}{7788 - 733} = 12.76 \text{ pF/tick}, \quad b = 10000 - 12.76 \times 733 = +650 \text{ pF},$$

equivalently $K = aR \approx 1.267 \times 10^6$ and $b = 650 \text{ pF}$ in Equation 8.

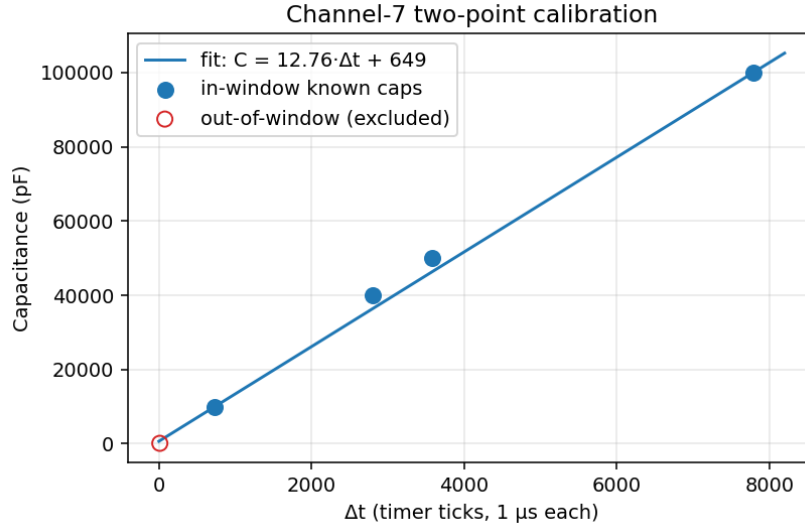


Figure 1: Channel-7 calibration. The in-window points (filled) define the affine fit; the out-of-window 100 pF point (open) illustrates why such points are excluded. The non-zero intercept is the offset b .

The figure was produced by the two-point fit below (run separately; the report embeds the resulting image so it renders without a Jupyter backend):

```
import numpy as np, matplotlib.pyplot as plt
data = [(733,10000),(3579,50000),(7788,100000),(2802,40000)] # in-window (Δt, C_pF)
a = (100000-10000)/(7788-733) # = 12.76 pF/tick
b = 10000 - a*733 # = 650 pF
# C = a·Δt + b (see eq. for the affine calibration model)
```

6.2 Automatic-ranging across seven decades

With ranging enabled and the channel-7 calibration applied, capacitors from 100 pF to 470 μF were measured (Table 2). The instrument selected an appropriate channel in every case.

Table 2: Auto-ranging results. The selection logic is correct throughout; accuracy is excellent wherever the multiplexer resistance is negligible.

Capacitor	Channel	Reading	Comment
0.33 μF	R3 (980 Ω)	0.337 μF	accurate
1 μF	R5 (9.86 kΩ)	0.975 μF	accurate
2.2 μF	R5	2.07 μF	within electrolytic tolerance
10 μF	R4 (5.5 kΩ)	8.76 μF	within electrolytic tolerance
47 μF	R3	46.5 μF	accurate
220 μF	R1 (56 Ω)	715 μF	gross error — see §7
680 μF	R0 (10 Ω)	overflow	beyond range — see §7

7 Discussion

7.1 Multiplexer on-resistance

The two failures at the smallest resistors share a single cause: the analog multiplexer’s channel **on-resistance** R_{on} , which adds in series with the selected ladder resistor. The true charging resistance is $R_{\text{eff}} = R_{\text{nom}} + R_{\text{on}}$, and since calibration used the nominal value, low-Ω ranges over-read.

We can extract R_{on} from the data. Treating the 220 μF capacitor as nominal and inverting Equation 5,

$$R_{\text{eff}} = \frac{\Delta t}{C \ln 2} = \frac{31617 \mu\text{s}}{(220 \mu\text{F})(0.693)} \approx 207 \Omega,$$

so on channel R1 ($R_{\text{nom}} = 56 \Omega$) the multiplexer contributes $R_{\text{on}} \approx 150 \Omega$ — entirely consistent with the CD74HC4067 datasheet. The resulting error scales as $(R_{\text{nom}} + R_{\text{on}})/R_{\text{nom}}$:

Channel	R_{nom}	R_{eff}	uncorrected error
R0	10 Ω	160 Ω	16×
R1	56 Ω	206 Ω	3.7×
R3	980 Ω	1130 Ω	+15%
R4	5.5 k Ω	5.65 k Ω	+2.7%
R7	99.3 k Ω	99.45 k Ω	+0.15%

This table explains everything observed: the gross 3.7× over-reading at 220 μF (R1), the impossibility of measuring 680 μF (R_on raises R_{eff} until the shortest available time constant still overflows the timer), and equally *why channel 7 calibrated so cleanly* — there the 150 Ω contributes a negligible 0.15%. The remedy is to populate the resistor table with **measured effective resistances** (the directly measurable pin-to-node resistance with each channel enabled) and, for the largest capacitors, to select a coarser timer prescaler to extend the upper count limit.

7.2 Random scatter and dielectric effects

Individual cycles scatter by $\pm 15\%$ because the comparator, lacking hysteresis, dithers at the slow crossing; the reported **trimmed mean** of 30 cycles (dropping the three highest and lowest) and the median agree to a few percent, confirming a stable central estimate. Notably, the instrument is *honest* about the components: the “104” ceramic, nominally 100 nF, measured 40 nF — matching a bench multimeter — because Class II ceramics lose capacitance under DC bias, whereas the film and electrolytic parts read within tolerance. The meter reports the *actual* capacitance at the test conditions, not the nameplate value.

8 Calibration simulation

To generalize the calibration across all ranges, the measurement was modelled in software as

$$\Delta t = p R_{\text{eff}} (C + C_{\text{stray}}), \quad R_{\text{eff}} = R_{\text{nom}} + R_{\text{on}}, \quad (9)$$

with $p = \ln(\text{ratio})/t_{\text{tick}}$ absorbing the clock and threshold-ratio errors. This is precisely the firmware’s own relation $C = \Delta t K / R + b$ inverted, with $K = 1/p$ and $b = -C_{\text{stray}}$ — so the simulation is structurally identical to the instrument’s algorithm. Given a trial R_{on} the model is linear in $(p, p C_{\text{stray}})$, fitted by weighted least squares (film references weighted above $\pm 20\%$ electrolytics).

The fit recovers a multiplexer resistance $R_{\text{on}} \approx 75\text{--}150 \Omega$ (consistent with the direct estimate of §7.1), a slope correction of 1.08 (within the internal oscillator’s $\pm 10\%$), and a stray offset of $\approx 600 \text{ pF}$ that independently reproduces the C7 two-point intercept — confirming the offset is a genuine, range-independent stray. Figure 2 shows the eleven channels tiling roughly 47 pF–900 μF with healthy overlap; Figure 3 shows the modelled versus measured intervals.

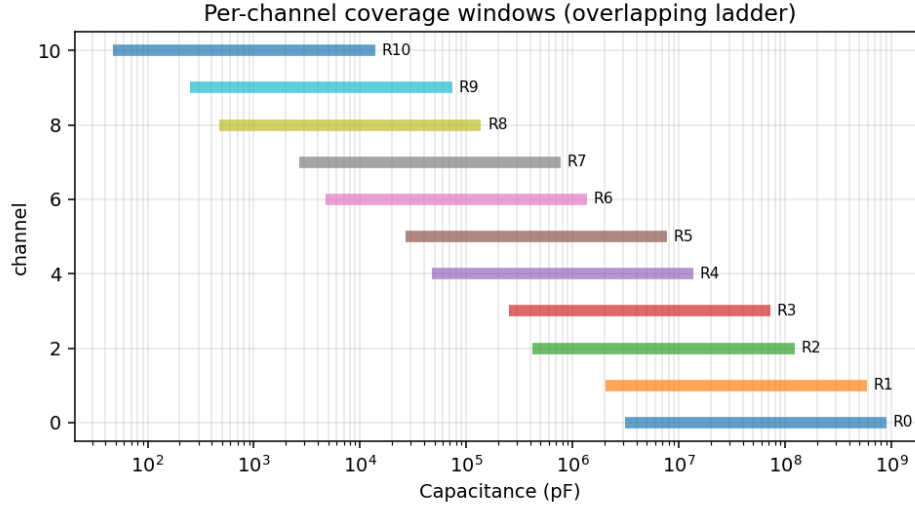


Figure 2: Per-channel coverage windows. Each resistor admits ~ 2.5 decades; the geometric ladder overlaps to span seven decades.

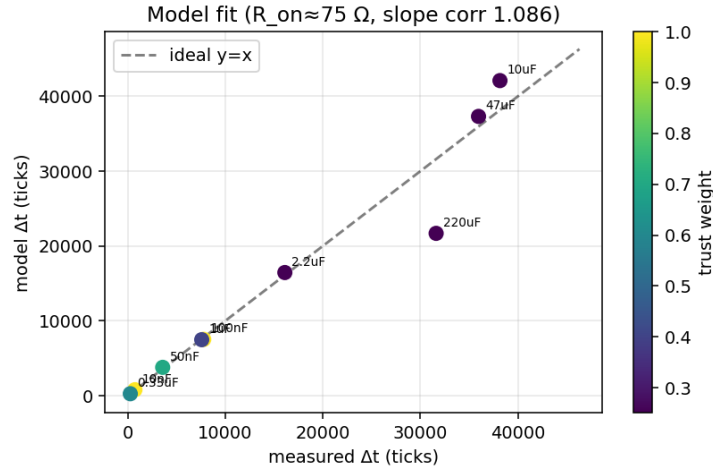


Figure 3: Modelled vs measured Δt , coloured by reference trust. Mid-range (film) points fit well; the low-trust electrolytic extremes ($\pm 20\%$) scatter — the limit on a precise cross-range fit.

The simulation's value is therefore twofold: it validates the firmware equation and resistor ladder, and it isolates the two remaining error sources — multiplexer resistance and an under-determined offset. A definitive universal calibration awaits **film reference capacitors measured on the low-resistance channels**, which the $\pm 20\%$ electrolytics used here cannot provide.

9 Conclusions and future work

We have realized a supply-independent, auto-ranging capacitance meter on a modest 8-bit microcontroller, validated from 100 pF to 470 μ F. The two-threshold timing method cleanly cancels the supply voltage (Equation 4), the bounded-counter and time-constant arguments together justify automatic ranging, and a two-point affine calibration (Equation 6) corrects the residual systematic errors. The dominant remaining limitation — multiplexer on-resistance at low ranges — has been identified and quantified at $R_{on} \approx 150 \Omega$.

Three refinements are proposed. First, replacing the nominal ladder values with **measured effective resistances** will recover accuracy on the low- Ω channels. Second, a **per-range timer prescaler** will extend the upper limit to the millifarad region. Third, a **multi-point least-squares calibration** across ranges — the basis of the planned simulation study — will distinguish a constant stray-capacitance offset from a resistance-dependent timing offset and thereby deliver uniform accuracy across all seven decades.